

五次Hermite插值多项式（基函数法）

给定节点数据：

x	0	1	2
$f(x)$	0	2	1
$f'(x)$	1	0	-2

构造五次Hermite插值多项式 $H_5(x)$ 满足
 $H_5(x_i) = f(x_i), H_5'(x_i) = f'(x_i), i = 0, 1, 2.$

1. 拉格朗日基函数

$$l_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{1}{2}(x^2 - 3x + 2),$$

$$l_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2) = -x^2 + 2x,$$

$$l_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{1}{2}x(x-1).$$

2. Hermite基函数

$$h_i(x) = \left[1 - 2(x-x_i) \sum_{\substack{j=0 \\ j \neq i}}^2 \frac{1}{x_i - x_j} \right] [l_i(x)]^2, \quad g_i(x) = (x-x_i)[l_i(x)]^2.$$

- 对于 $i = 0$:
$$\sum_{j \neq 0} \frac{1}{x_0 - x_j} = \frac{1}{0-1} + \frac{1}{0-2} = -1 - \frac{1}{2} = -\frac{3}{2},$$

故 $1 - 2(x-x_0) \cdot \left(-\frac{3}{2}\right) = 1 + 3x.$
- 对于 $i = 1$:
$$\sum_{j \neq 1} \frac{1}{x_1 - x_j} = \frac{1}{1-0} + \frac{1}{1-2} = 1 - 1 = 0,$$

故 $1 - 2(x-x_1) \cdot 0 = 1.$

- 对于 $i = 2$:

$$\sum_{j \neq 2} \frac{1}{x_2 - x_j} = \frac{1}{2-0} + \frac{1}{2-1} = \frac{1}{2} + 1 = \frac{3}{2},$$

$$\text{故 } 1 - 2(x - x_2) \cdot \frac{3}{2} = 1 - 3(x - 2) = 7 - 3x.$$

于是得到Hermite基函数:

$$h_0(x) = (1 + 3x)[l_0(x)]^2 = (1 + 3x) \cdot \frac{1}{4}(x^2 - 3x + 2)^2,$$

$$g_0(x) = (x - 0)[l_0(x)]^2 = x \cdot \frac{1}{4}(x^2 - 3x + 2)^2,$$

$$h_1(x) = 1 \cdot [l_1(x)]^2 = x^2(x - 2)^2,$$

$$g_1(x) = (x - 1)[l_1(x)]^2 = (x - 1)x^2(x - 2)^2,$$

$$h_2(x) = (7 - 3x)[l_2(x)]^2 = (7 - 3x) \cdot \frac{1}{4}x^2(x - 1)^2,$$

$$g_2(x) = (x - 2)[l_2(x)]^2 = (x - 2) \cdot \frac{1}{4}x^2(x - 1)^2.$$

3. 插值多项式

$$\begin{aligned} H_5(x) &= \sum_{i=0}^2 f(x_i)h_i(x) + \sum_{i=0}^2 f'(x_i)g_i(x) \\ &= 0 \cdot h_0(x) + 2 \cdot h_1(x) + 1 \cdot h_2(x) + 1 \cdot g_0(x) + 0 \cdot g_1(x) + (-2) \cdot g_2(x) \\ &= 2x^2(x - 2)^2 + \frac{7 - 3x}{4}x^2(x - 1)^2 + \frac{x}{4}(x^2 - 3x + 2)^2 - \frac{1}{2}x^2(x - 1)^2(x - 2). \end{aligned}$$

合并得:

$$H_5(x) = 2x^2(x - 2)^2 + \frac{11 - 5x}{4}x^2(x - 1)^2 + \frac{x}{4}(x - 1)^2(x - 2)^2.$$

$$\begin{aligned} 2x^2(x - 2)^2 &= 2x^4 - 8x^3 + 8x^2, \\ \frac{11 - 5x}{4}x^2(x - 1)^2 &= -\frac{5}{4}x^5 + \frac{21}{4}x^4 - \frac{27}{4}x^3 + \frac{11}{4}x^2, \\ \frac{x}{4}(x - 1)^2(x - 2)^2 &= \frac{1}{4}x^5 - \frac{3}{2}x^4 + \frac{13}{4}x^3 - 3x^2 + x. \end{aligned}$$

相加得:

$$H_5(x) = -x^5 + \frac{23}{4}x^4 - \frac{23}{2}x^3 + \frac{31}{4}x^2 + x.$$

4. 计算 $H_5(1.5)$

$$\begin{aligned}H_5(1.5) &= -(1.5)^5 + \frac{23}{4}(1.5)^4 - \frac{23}{2}(1.5)^3 + \frac{31}{4}(1.5)^2 + 1.5 \\&= -7.59375 + 5.75 \times 5.0625 - 11.5 \times 3.375 + 7.75 \times 2.25 + 1.5 \\&= -7.59375 + 29.109375 - 38.8125 + 17.4375 + 1.5 \\&= 1.640625.\end{aligned}$$

因此，五次Hermite插值多项式为 $H_5(x) = -x^5 + \frac{23}{4}x^4 - \frac{23}{2}x^3 + \frac{31}{4}x^2 + x$ ，且 $H_5(1.5) = 1.640625$ 。